

E2 — independent mathematics (human-style rederivation by the AI auditor)

Obligation. Reconstruct the proof-critical arguments independently; identify every step known only through Lean or an imported paper and whether the prose suffices for human review.

Input. EN md 71417794... (full read, 11:01–11:05 –03:00). No internal-audit verdict or derivation was read before this record (reading order §1.2). Method: line-by-line rederivation from the stated hypotheses; exact numerics delegated to E5 (e5_exact.py, 74 checks, 0 failures) and bounded searches to E5 (e5_search.py).

Labels: `CALCULATION_VERIFIED` (auditor rederived), `COMPUTATION_VERIFIED` (E5 exact), `LEAN_ONLY` (prose is a sketch; the auditor could not rederive it from the text alone; correctness rests on the formal proof), `IMPORTED` (proof in a prior paper; not re-audited here).

#	Component (EN location)	Status	Auditor notes								
1	Gain/loss identity, Lemma 2.1, (1.3), (1.3a)	CALCULATION_VERIFIED		Q	= e – g(P); for any partition g(Q)+	Q	= e with ordinary gain C(	K	,2)–1.		
2	LP duality, (2.2)–(2.4); weight 5/6 bound	CALCULATION_VERIFIED	$g(K) \leq (5/6)$	E(K)	for K_3 ($2 \leq 5/2$) and K_4 ($5 = 5$). Strong duality: IMPORTED (Paper I, Fourier–Motzkin in Lean).						
3	(2.5) and $\lfloor (2n+1)^2/24 \rfloor = M(n)$	CALCULATION_VERIFIED + COMPUTATION_VERIFIED	$n(n+1) \equiv 0,2 \pmod{6}$.								
4	Theorem 3.1 (RC01) as a whole	LEAN_ONLY / IMPORTED	Lemma 3.2 (bounded-rank nibble) IMPORTED from Paper III.								
4a	Lemma 3.3 (two quotas via auxiliary vertices)	CALCULATION_VERIFIED	Both quota inequalities rederived; uses $r(u+a) \leq$	U	(r-uniform) and $2b \leq \beta, \varepsilon$. Codegree of original–auxiliary pairs $\leq 1/p \leq \gamma_0$.						
4b	Lemma 3.4 (pairing of disjoint triangles, marks for K_4)	CALCULATION_VERIFIED	Pair mass $\geq (t_3-3)/2$; edge loads do not increase; cross-member codegree $\leq 1/t_3$ (the text's $3/t_3$ is conservative); gain $4m + b_4 \geq (1-\beta)(2t_3-6+5t_4) - 5\varepsilon$	U	– 5C. Unique decomposition of a 6-edge union of two edge-disjoint triangles checked by hand.						
4c	C.1–C.2 cleaning accounts (3.7), (3.10) charging 3 per discarded edge, $K_4 \rightarrow$ faces at $x/2$	CALCULATION_VERIFIED (local charging rules)	The counting of discarded edges $(3\delta + 1/k_0 + d)n^2$ itself depends on the regularity partition and exceptional-class conventions that are not written out: LEAN_ONLY.								
4d	Joint codegree (3.12)–(3.13), $\psi_H \leq t^2$ normalization	CALCULATION_VERIFIED (per-profile bound $1/(a_3t)$, $1/(a_4t)$)	Fibre lower bounds $b_H \geq a_3t^3, a_4t^4$ (counting lemma for regular pairs) and (3.8) retention: LEAN_ONLY.								
4e	C.3 schedule: a, δ , z, k_0 , h, L_T, M_T	COMPUTATION_VERIFIED	$h = \lfloor 4/(\delta/8)^5 \rfloor$ with $\delta = a^{21}/2208$, matching Mathlib's SzemerédiRegularity iteration count convention (E3). Table C.1 rows $\gamma_T \geq 2^{59485}$ etc. and the domination "after two iterations": LEAN_ONLY (only asserted).								
5	Corollary 3.5	CALCULATION_VERIFIED	Needs $M(n) \geq n^2/6$ for $n \geq 6$ (E5; fails for $n < 6$ as negative control).								

6	Lemma 4.1: mixed transport (4.4), twin potential, existence of a non-twin simplicial pair outside the complete-split family, chordality preservation	CALCULATION_VERIFIED	Transport: loads ≤ 2 after mapping both copies, average restores capacity; $e(G_{uv})+e(G_{vu}) = 2e(G)$. Existence: rederived with the rooted Dirac argument (component with an edge contains a simplicial vertex of G). Lex potential (F_4 , twin pairs) strictly increases; $+2(s-t+1) > 0$. Random tests in E5.		
7	Lemma 4.2 and descent (4.5)–(4.6a); no circularity	CALCULATION_VERIFIED + COMPUTATION_VERIFIED	Theorem 5.0's proof (Lemma 5.1 onward) uses only $d(G) < \varepsilon_0$ and the value hypothesis, not Prop 4.3. F_4 monotone along path preserves the value hypothesis. Crossing point of (4.6a) $\approx 3.6725 \cdot 10^{12}$ (E5), consistent with "fails for $n \leq 3.6 \cdot 10^{12}$ ".		
8	Prop 4.3 (4.7)–(4.8)	CALCULATION_VERIFIED	From descent $d(G) < \varepsilon_0 - 1/n$ and Lemma 5.1's comparator estimate.		
9	Lemma 5.1: (5.1a) three regimes, completing the square, PEO edge bound $(a-1)r - C(a,2)$, missing pairs $C(u+1,2)$, window constants	CALCULATION_VERIFIED + COMPUTATION_VERIFIED	Robustness	ΔF_4	≤ 4 per edit (the text's 6 is conservative). Fractional realizability of the symmetric K_3/K_4 mixtures in (5.1a) by symmetry: AUDITOR_INFERENCE (plausible, Lean-checked per A.2).
10	Prop 5.2: R is a clique; (5.3); size bounds $p/a, q/a, 48(2p-q)_+ \leq a, 3D \leq a$	CALCULATION_VERIFIED	Common-neighbourhood-is-a-clique argument rederived.		
10a	(5.4) mass bounds $400m < a^2$, $2000(A+2m) \leq 11a^2$, and palette/width (5.7) $40c \leq 73p, 40(w-1) \leq 3p$, and the degree/width inequalities before (5.4)	LEAN_ONLY (orders of magnitude checked)	The text says "Substituting ... gives (5.4) and (5.7)" without the computation; the degree bound $64d_{\max} \leq 113a + 64$	X	is asserted. A human reader cannot verify (5.7) from the prose.
11	Table 2 piece identity (5.4a) and compatibility	CALCULATION_VERIFIED			
12	First phase (5.5)–(5.7) $\rightarrow 1600m \leq 2920f + 219A$	CALCULATION_VERIFIED given (5.7)	Equitable recolouring (Kempe), PEO ending in R orientation, cyclic shifts averaging.		
13	Second phase (5.8)–(5.14)	CALCULATION_VERIFIED + COMPUTATION_VERIFIED	Moments, failure probabilities, linearization; all rational comparisons exact (E5).		
14	Lemma 5.3 coefficients, (5.16), $B_n(p) \leq M(n)$	CALCULATION_VERIFIED + COMPUTATION_VERIFIED	Substitution of the lower bound on f is valid because its coefficient is negative (independent of the sign of the bound).		

15	Cor 5.4 (5.17)–(5.18)	CALCULATION_VERIFIED + COMPUTATION_VERIFIED			
16	Assembly of Theorem B; thresholds fixed in advance	CALCULATION_VERIFIED			
17	§6.1 unrestricted lower bound, (6.2); §6.2 (6.3)–(6.4); §6.4 (6.6)	CALCULATION_VERIFIED + COMPUTATION_VERIFIED	ILP confirmation for $2 \leq k \leq 5$, $k \leq h \leq 7$ (E5).		
18	Theorem A all orders, $b =$ N^2 , tower propagation; optimality of $1/6$ (quadratic and linear)	CALCULATION_VERIFIED	Tower internals LEAN_ONLY (item 4e).		
19	Theorem 6.1: $\gamma = \eta_0/4$, far branch excluded, (6.7) from (5.15a), 16δ	CALCULATION_VERIFIED	Far branch:	Q	$< M - \eta_0 n^2/2 < M - \delta$. 16 is exactly the best constant derivable from the retained account (6.7) (LP in E5: max $m +$ $A = 16\delta$).
20	Corollary 6.1a: (6.7a), (6.7b), 48δ , quantifier order	CALCULATION_VERIFIED + COMPUTATION_VERIFIED	Identity (6.7b) rederived (edges internal/exterior/cross counted exactly) and tested on exact optimal partitions (E5). 48 is tight for the account (6.7) (LP max $r + A +$ $3m = 48\delta$; 47 fails — negative control). With the stronger (5.15a) coefficients one could obtain $\approx 46.8\delta$; not a defect. τ ≥ 0 not needed (Lean has $\tau \in$ $\mathbb{Q}$). Root chosen before Q.		
21	Cor 6.2 classification; Prop 6.3; (6.12)	CALCULATION_VERIFIED + COMPUTATION_VERIFIED			
22	Prop D.1 table / D.2	CALCULATION_VERIFIED (arithmetic) ; constructions IMPORTED/LEAN_ONLY (Bose/Skolem frames)	ILP: $c_4(K_n) \leq M(n)$ for $n \leq 10$; $c_3(K_{10}) \geq 19$ (E5).		
23	Theorem C: E.2 induction removing K_s ; E.3 witness and unrestricted lower bound	CALCULATION_VERIFIED + COMPUTATION_VERIFIED	(E.3) increments hold for all n ≥ 1 ; witness $rsd \leq s$ by hand and by brute force for small n (E5); (E.5) identity.		
23a	E.1 normalization bounds ($r \leq$ $4p/10^{27}$ etc.), (E.2b), (E.2d), three-case budget	LEAN_ONLY	The prose gives the architecture (IndepAllPeel/ Obstr/Budget) but not the derivations; not human- verifiable from the text. (E.2a) and (E.2c) $\rightarrow$ (E.2e) $\rightarrow$ $Q_s(n)$ CALCULATION_VERIFIED.		
24	§8.2 (8.1a), §8.3 Prop 8.1	CALCULATION_VERIFIED (8.3)–(8.5) and the Behrend/ Ruzsa–Szemerédi arithmetic	Mathlib's <code>roth_lower_bound</code> statement not re-read here.		

Negative controls. See E5 (5 exact + 4 search controls). In E2 itself: the auditor checked that replacing $M(n)$ by the rational envelope $(2n+1)^2/24$ breaks the integrality step $(5.16) \rightarrow B \leq M$ (E5 negative control), and that the far-branch exclusion in Thm 6.1 fails if $\gamma > \eta_0/2$ (then δ could exceed $\eta_0 n^2/2$): with $\gamma = \eta_0/4$ there is slack, as written.

No mathematical defect found in the rederived parts. **Limitation (material for the verdict):** items 4, 4c, 4d, 4e, 10a, 22 (constructions) and 23a are not human-reviewable from the manuscript prose alone; their correctness rests on the Lean development, whose dynamic check (E4) has **not** been run in this audit (stopped by the E7 blocker). **E2 = INCONCLUSIVE** (no defect found; coverage incomplete for LEAN_ONLY steps).

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